

Algebra 1: Polynomials

Challenge Level

Name: _____

Date: _____

Question 1: Simplify the expression $x^2 + 3x + 2$ by factoring.

- A) $x(x + 1)$
- B) $x(x + 2)$
- C) $(x + 1)(x + 2)$
- D) $(x - 1)(x - 2)$
- E) $(x + 1)(x - 2)$

Question 2: Solve the equation $2x^2 + 5x - 3 = 0$.

- A) $x = -3$ or $x =$

$\frac{3}{2}$

- A) $x = -1$ or $x = 3$
- B) $x = 1$ or $x = -3$
- C) $x = -$
 $\frac{3}{2}$ or $x = 3$
- C) $x = 3$ or $x = -3$

Question 3: Find the value of x in the equation $x^2 - 7x + 12 = 0$.

- A) $x = 3$ or $x = 4$
- B) $x = 2$ or $x = 5$
- C) $x = 3$ or $x = 5$
- D) $x = 4$ or $x = 6$
- E) $x = 2$ or $x = 6$

Question 4: Determine the equation of the polynomial that has zeros at $x = -2$, $x = 1$, and $x = 3$.

- A) $y = (x + 2)(x - 1)(x - 3)$
- B) $y = (x - 2)(x + 1)(x + 3)$
- C) $y = (x + 2)(x + 1)(x + 3)$
- D) $y = (x - 2)(x - 1)(x - 3)$
- E) $y = (x + 2)(x - 1)(x + 3)$

Question 5: Simplify the expression $(2x^2 + 3x - 1) + (x^2 - 2x - 3)$.

- A) $3x^2 + x - 4$
- B) $3x^2 + x - 2$

- C) $3x^2 - x - 4$
- D) $x^2 + x - 2$
- E) $x^2 - x - 4$

Question 6: Solve the inequality $x^2 - 4x - 3 > 0$.

- A) $x < -1$ or $x > 3$
- B) $x > -1$ or $x < 3$
- C) $x > -1$ and $x < 3$
- D) $x < -1$ and $x > 3$
- E) $x = -1$ or $x = 3$

Question 7: Factor the polynomial $x^3 - 2x^2 - 5x + 6$.

- A) $(x - 1)(x - 2)(x + 3)$
- B) $(x + 1)(x - 2)(x + 3)$
- C) $(x - 1)(x + 2)(x - 3)$
- D) $(x - 1)(x - 2)(x - 3)$
- E) $(x + 1)(x + 2)(x + 3)$

Question 8: Find the value of x in the equation $x^2 + 2x - 6 = 0$.

- A) $x = -1 - \sqrt{7}$ or $x = -1 + \sqrt{7}$
- B) $x = -3$ or $x = 2$
- C) $x = -1 - \sqrt{5}$ or $x = -1 + \sqrt{5}$
- D) $x = -2$ or $x = 3$
- E) $x = 1$ or $x = -6$

Question 9: Show all the steps to simplify the expression $(x^2 + 2x + 1)(x^2 - 2x + 1)$ and then factor the resulting expression.

Question 10: Prove that $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$ by showing all the steps of the algebraic manipulation, starting with the right-hand side of the equation.

Chef's Tips (Hints):

- To factor polynomials, look for common factors and use techniques such as grouping and the difference of squares.
- To solve quadratic equations, use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, and to solve inequalities, use sign analysis and test points in each interval.
- When simplifying expressions, combine like terms, and when factoring, look for patterns that fit known factorable forms.